19 JANUARY 2038 INDIA

FAKE NEWS DETECTION

NEWS UPDATE

USING DISTILBERT

The rapid growth of social media and digital news platforms has made information easily accessible. However, it has also increased the spread of fake news, misinformation, and misleading content. Fake news can influence public opinion, create social unrest, and affect decision-making processes.

Manual verification of news articles is time-consuming and often impractical due to the huge volume of content generated daily. Therefore, an automated system capable of distinguishing between real and fake news is essential.

In this project, a Natural Language Processing (NLP) based Fake News Detection System was developed using DistilBERT, a transformer-based deep learning model from Hugging Face. The model was trained on approximately 45,000 news articles and achieved 99.60% classification accuracy. The solution was further deployed as a web application using Streamlit Cloud, with the trained model hosted on Hugging Face Hub.

INTERNSHIP PROJECT PRESENTATION
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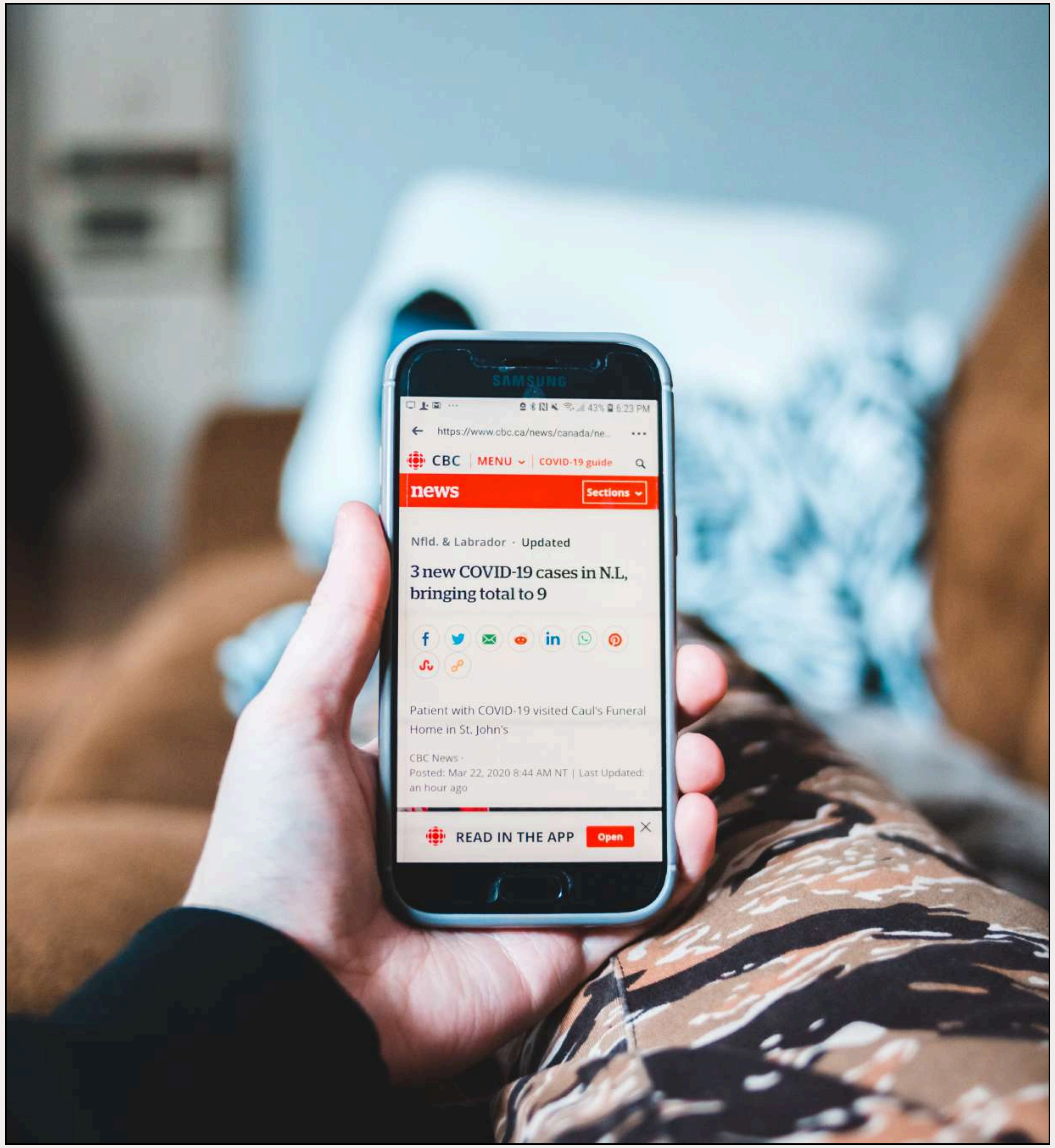
PROBLEM STATEMENT

- The rapid growth of social media and online news platforms has increased the spread of fake news and misinformation.
- Fake news can mislead people, influence public opinion, and create social and political issues.
- Manual verification of news articles is time-consuming and difficult due to the large volume of content generated daily.
- Traditional methods often fail to understand the contextual meaning of news articles.
- Therefore, an automated and intelligent system is needed to accurately identify fake news.
- This project uses DistilBERT, a transformer-based NLP model, to classify news articles as Real or Fake.
- The goal is to provide a fast, accurate, and scalable solution for fake news detection.



DATASET

- The dataset was collected from the Kaggle Fake and Real News Dataset.
- It contains news articles categorized as Fake News and Real News.
- The dataset includes important features such as Title, Text, Subject, and Date.
- Total records: 44,898 news articles (23,481 Fake and 21,417 Real).
- Duplicate records were removed, resulting in 44,689 unique articles.
- The Title and Text columns were combined to create the final content used for training.
- This dataset was used to train and evaluate the DistilBERT model for fake news classification.

WORKFLOW

🔄 Workflow of Fake News Detection System

Data Collection

Collected Fake and Real news articles from the Kaggle dataset.

Data Preprocessing

Combined Title and Text columns.
Converted text to lowercase.
Removed URLs, HTML tags, special characters, and extra spaces.

Data Preparation

Assigned labels: Fake = 1, Real = 0.
Split dataset into Training and Testing sets.

Tokenization

Used DistilBERT Tokenizer to convert text into numerical tokens understandable by the model.

Model Training

Fine-tuned the pre-trained DistilBERT model on the news dataset.
Learned contextual and semantic relationships within the text.

Model Evaluation

Evaluated using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.
Achieved 99.60% Accuracy.

Model Saving

Saved the trained DistilBERT model and tokenizer for future predictions.

Deployment

Uploaded the model to Hugging Face Hub.
Developed a Streamlit web application for real-time fake news prediction.
Deployed the application on Streamlit Cloud.

Workflow Diagram

Kaggle Dataset
↓
Data Preprocessing
↓
Label Encoding
↓
Train-Test Split
↓
DistilBERT Tokenization
↓
Model Training
↓
Model Evaluation
↓
Save Trained Model
↓
Hugging Face Hub
↓
Streamlit Web App
↓
Fake / Real Prediction

TOP STORY

Literature brings us closer to unexplored worlds and deep emotions that resonate in the human soul.

Model Training

- DistilBERT, a lightweight version of BERT, was selected for fake news classification due to its speed and efficiency.
- News articles were tokenized using the DistilBERT tokenizer with a maximum sequence length of 256 tokens.
- The model was fine-tuned on the prepared dataset using 1 training epoch.
- A batch size of 4 was used to optimize memory usage and training performance.
- After training, the model achieved 99.60% accuracy, demonstrating excellent performance in distinguishing real and fake news.

Results & Evaluation

The trained DistilBERT model was evaluated on the test dataset.
The model achieved an impressive 99.60% accuracy in classifying news articles.
Precision, Recall, and F1-Score values were close to 1.00, indicating highly reliable predictions.
The Confusion Matrix showed only 8 misclassifications out of 2000 test samples.
These results demonstrate the effectiveness of DistilBERT in understanding the context and identifying fake news accurately.

SPECIAL
BROADCAST

Deployment & Application

- The trained DistilBERT model was uploaded to the Hugging Face Hub for easy access and model management.

- A user-friendly web application was developed using Streamlit.

- Users can enter a news article and instantly receive a prediction as Real News or Fake News.

- The application also displays a confidence score to indicate prediction reliability.

- The complete solution was deployed on Streamlit Cloud, making it accessible online from anywhere.

DEPLOYMENT ARCHITECTURE

GitHub
↓
Hugging Face Model Hub
↓
Streamlit Cloud
↓
User Prediction Interface

Conclusion

- A Fake News Detection System was successfully developed using DistilBERT and NLP techniques.
- The model was trained on over 44,000 news articles and achieved 99.60% accuracy.
- The system can effectively classify news articles as Real or Fake.
- A Streamlit web application was developed and deployed for real-time predictions.
- The project demonstrates the potential of Transformer-based models in combating misinformation.

Future Scope

- Train the model on more diverse and recent news datasets.
- Extend support for multiple languages.
- Integrate real-time news feeds and fact-checking APIs.
- Improve performance on domains such as technology, science, and space news.

YEAR 2038

“THE YEAR 2038 PROBLEM OCCURS BECAUSE MANY OLDER SYSTEMS STORE UNIX TIME AS A SIGNED 32-BIT INTEGER. ON JANUARY 19, 2038, THE VALUE OVERFLOWS, CAUSING INCORRECT DATE CALCULATIONS. THE SOLUTION IS TO MIGRATE TO 64-BIT TIME REPRESENTATIONS AND UPDATE AFFECTED SOFTWARE AND DATABASES.“

THANK YOU

https://fake-news-detection-kasirajan.streamlit.app/